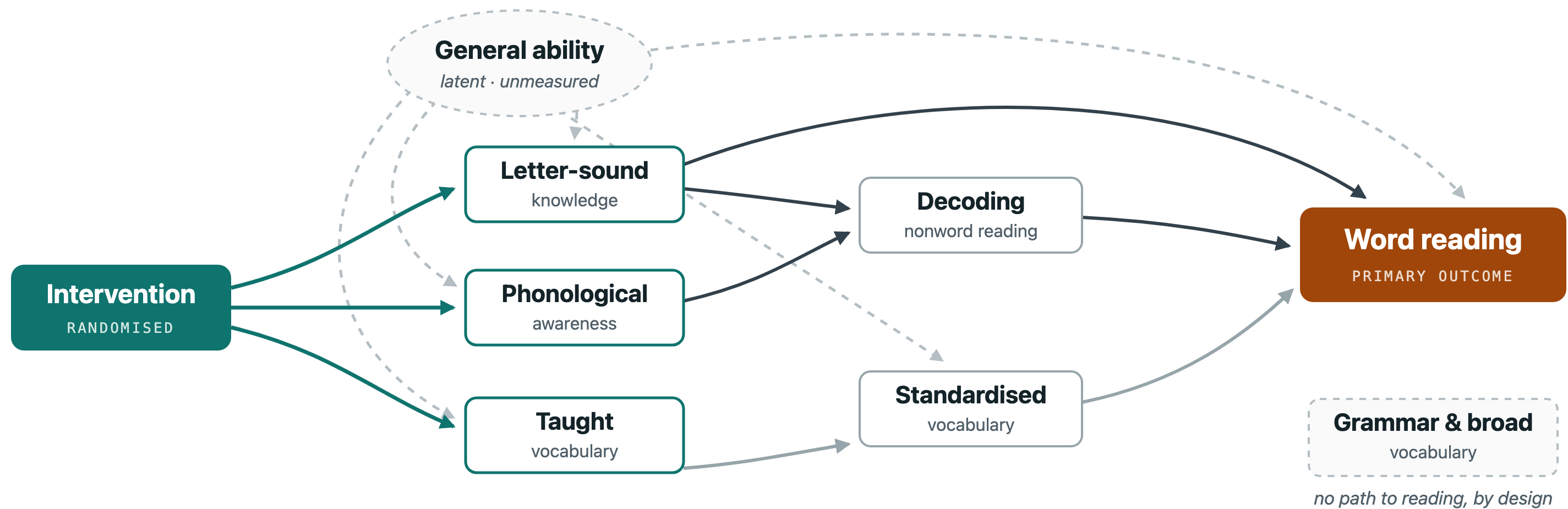


A causal design for reading progress

Separating what the teaching *causes* from what merely correlates with a child’s general ability.



HOW TO READ IT

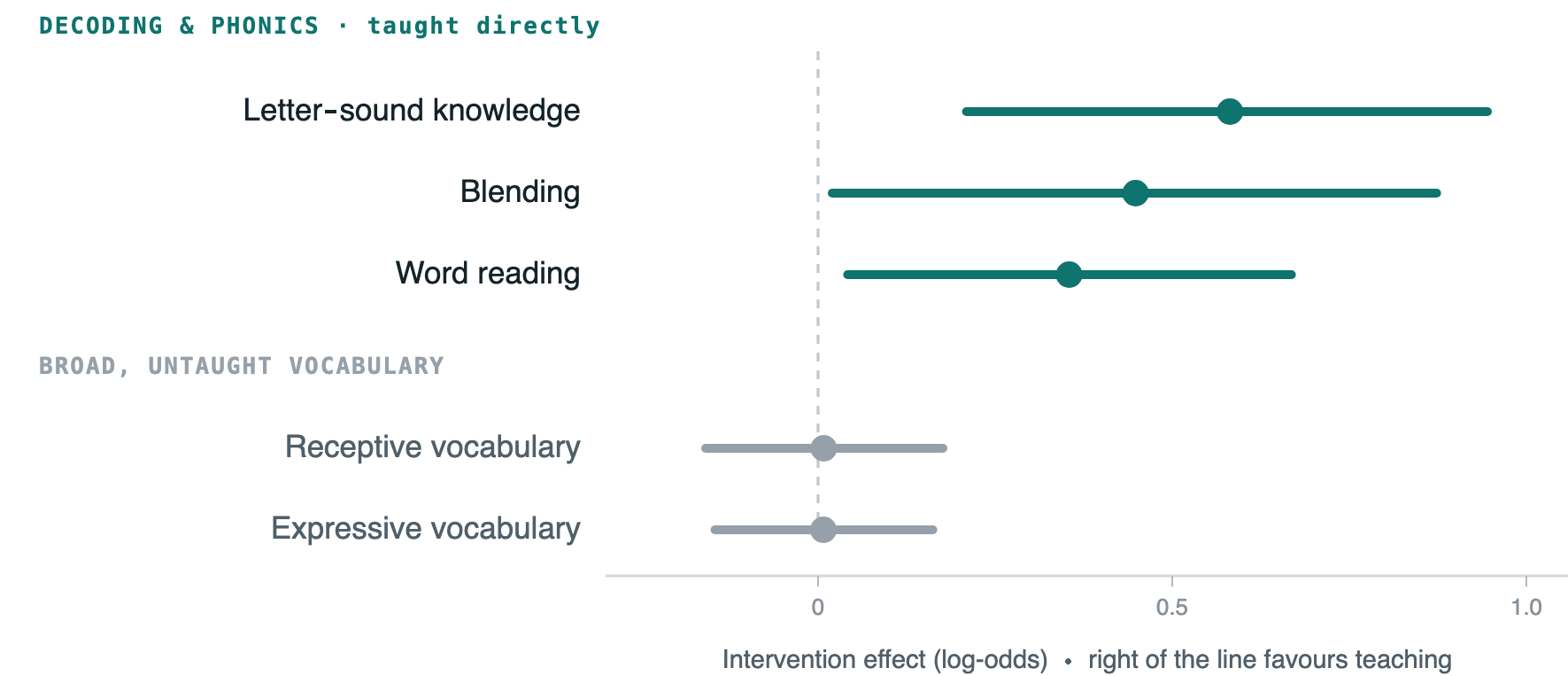
- Randomised effect, **causal**
- Hypothesised skill pathway, association
- - - - - General-ability confounding
- Association through standardised vocabulary
- Teal outline: a skill the teaching targets directly

THE IDENTIFICATION

Randomisation makes the intervention independent of general ability, so its effect on word reading is read as **genuine cause**. Every skill-to-skill link stays confounded by ability, so it is reported as **association, not cause**.

The intervention lifted reading, not broad vocabulary

The randomised effect of the teaching on each skill.



Each bar is the intervention's effect on one skill, with its **95% credible interval**. Children were randomised, so these read as cause, not correlation.

The gains cluster on the skills taught most directly: letter-sounds, blending and word reading. Broad, untaught vocabulary barely moved.

Learning transfers near what's taught, not far. The effect fades with distance from instruction. This is the transfer gradient.

● A bar clear of the dashed zero line is a credibly non-zero effect.